VICTORIA JUNIOR COLLEGE
BIOLOGY DEPARTMENT
JC2 PRELIMINARY EXAMINATIONS 2008
Higher 1

CANDIDATE NAME

CLASS

INDEX NUMBER

BIOLOGY 8875/02

Paper 2 Core Paper **9 September 2008**

Additional Materials: Answer Paper **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your CT GP/ INDEX no. and name on all the work you hand in.
Write in dark blue or blue pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use any staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions

Section B
Answer any **one** question.
Write your answers on separate answer paper provided.

At the end of the examinations,
1. hand in section A and the 2 questions you attempted from section B separately;
2. fasten all your work securely;
3. enter the number of the section B questions you have answered in the grid opposite.

The intended number of marks is given in brackets [ ] at the end of each question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
Section B	
Total	

This paper consists of **9** printed pages.

Section A: Structured questions
Answer all questions

Question 1

(a) **Figure 1** below shows a trans-membrane protein that is involved in ATP synthesis.

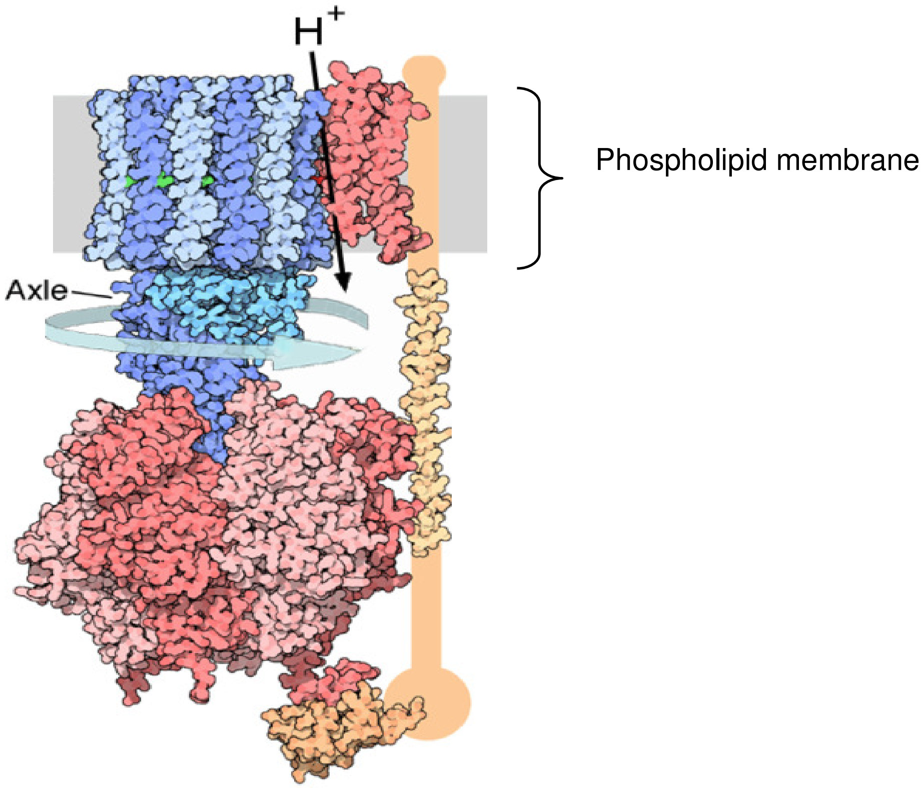

Figure 1

(i) State **two** specific locations in the cell organelles where this protein may be found.

Location 1……………………………… Location 2……………………………… [1]

(ii) With reference to the information provided, explain how the specific structure of the protein is determined.

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(b) The enzyme phenol oxidase is often released when plant cells are disrupted. This enzyme causes the oxidation of phenols into coloured products. In an experiment, samples of the enzyme extract were mixed with different substrate concentrations, with or without the presence of a certain chemical, PTU (phenylthiourea). The results are shown in the table below.

Substrate concentration / mmol dm^{-3}	**Initial rate with PTU / arbitrary units**	**Initial rate without PTU / arbitrary units**
0.5	2.4	4.2
1.0	4.1	6.3
1.5	5.1	7.1
2.0	5.5	7.6
2.5	5.5	7.6

(i) With reference to the table, suggest whether PTU acts as a competitive or non-competitive inhibitor of the enzyme.

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(ii) Explain your answer.

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…………………………………………………………………………………………...[2]

(c) Complete the table below to show the major differences between DNA and collagen. [4]

DNA	**Collagen**

[Total : 12 marks]

Question 2

(a) Outline briefly the structure of an amino acid and a dipeptide.

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(b) Describe how the structure of the ribosome allows it to perform its function in the cytoplasm. [3]

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…………………………………………………………………………………………[3]

(c) Explain how a single base mutation within a gene can result in a large change in the sequence the amino acids in a polypeptide chain when synthesized on a ribosome.

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(d) One of the features of the genetic code is that it is degenerate. Explain what that means. [2]

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(e) State 2 differences between prokaryotic and eukaryotic control of gene expression at the transcriptional level. [2]

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[Total : 13 marks]

Question 3

Fruit fly (*Drosophila melanogaster*) males are heterogametic. In a cross which was carried out between a female fly with normal wings and a male fly with 'cut' wings, all the offspring produced had normal wings.

In another cross, a female fly with cut wings was crossed with a male fly with normal wings. All the female offspring had normal wings BUT the male offspring had 'cut' wings.

(a) State and explain which trait is dominant.

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(b) Describe how these results show that the allele for wing shape is carried on the X- chromosome.

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Mutant flies with ebony trait have a much darker body colour as compared to the wild-type flies. In a separate cross, a female fly with normal body colour and normal wings, was crossed with a male fly with normal body colour and cut wings.

The following results were obtained:

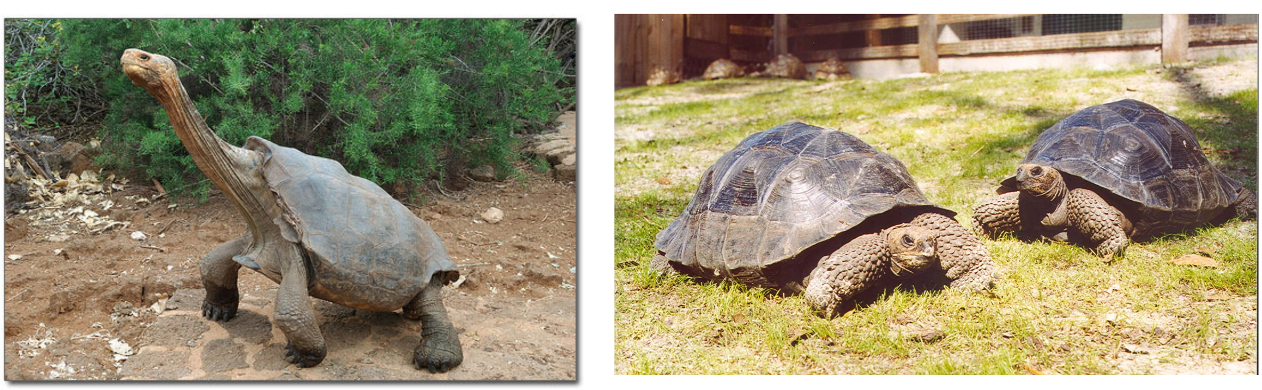
Offspring produced	**Number**
Female with normal body colour and normal wings	186
Female with normal body colour and cut wings	184
Female with ebony coloured body and normal wings	62
Female with ebony coloured body and cut wings	61
Male with normal body colour and normal wings	189
Male with normal body colour and cut wings	182
Male with ebony coloured body and normal wings	64
Male with ebony coloured body and cut wings	60

(b) Using appropriate symbols, draw a genetic diagram (in the space provided below and the next page) to explain the results of the cross. [6]

[Total: 12 marks]

Question 4

There are two main types of tortoises on the Galapagos Islands as shown in Figure 4 below. One has a domed shell and short neck and lives on the wetter islands while the other variety has a shell that allows its long neck to be raised and lives on the drier islands where the vegetation mainly consists of tall shrubs and bushes.



Long necked Short necked

Figure 4

(a) How would Darwin's theory of evolution explain the evolution of the long-necked form of the Galapagos tortoise?

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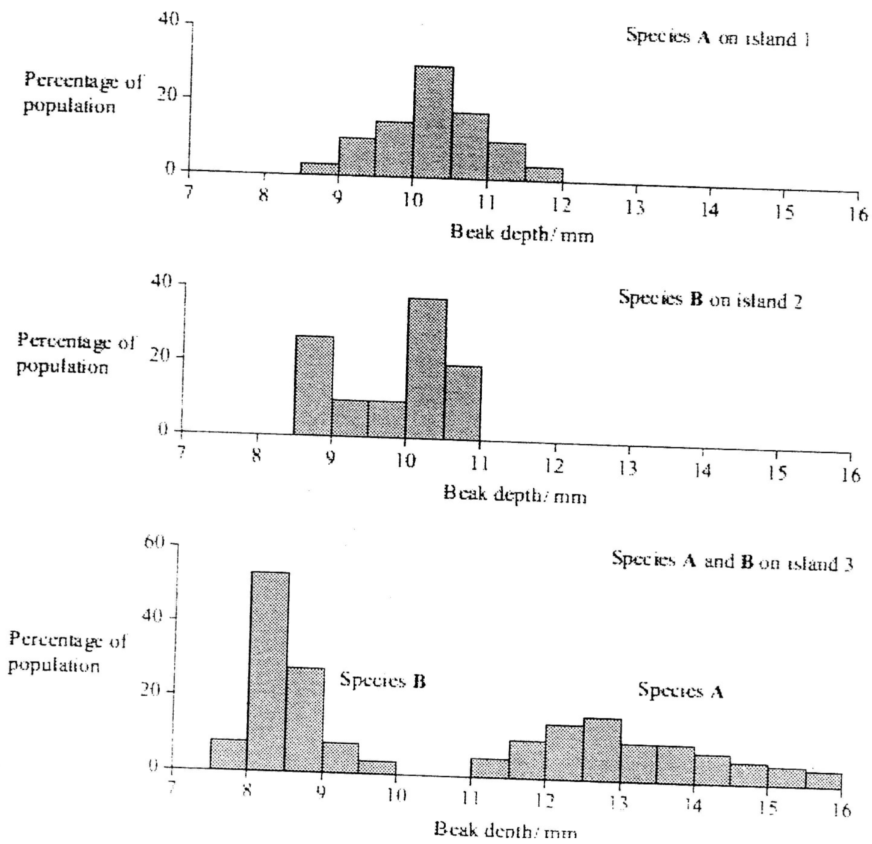
....[4]

Another group of scientists took measurements of the beak size of two species of finch (Species A and Species B) on different islands. Species A is found on island 1 while species B is found on island 2. Both species apparently can be found on island 3. They are thought to have colonised island 3 from islands 1 and 2 respectively. The figure below shows the graphs of the ranges of beak sizes of the two species on the different islands.



(b) State and explain the type of natural selection that took place in the populations of both finches after they have colonised island 3.

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(c) State and explain the difference between homologous and analogous structures.

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[Total: 10 marks]

ESQ

5(a). Explain the role of NAD and NADP in oxidative and photophosphorylation respectively. [8]

(b). Describe the main stages of Calvin Cycle and explain how radioactive carbon can be used to understand limiting factors on the rate of photosynthesis. [12]

6(a) Describe the properties of plasmids that would allow them to be use as vectors. [6]

(b) Explain the significance of genetic engineering in improving the quality of crop plants. Include a relevant example in your answer. [6]

(c) Discuss the goals and the benefits of the human genome project. [8]